

SETAREH NAJAFI

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EDUCATION

Ph.D. in Computer Science, University of California, Santa Barbara *GPA: 4.0/4.0* Jan. 2023 – 2028
B.Sc. in Computer Science, University of Tehran *GPA: 3.86/4.0* Sep. 2019 – July 2022

RESEARCH INTERESTS

Mechanistic Interpretability Machine Unlearning & Knowledge Removal in LLMs
Chain-of-Thought Reasoning Reinforcement Learning

RESEARCH EXPERIENCE

Research Assistant, UCSB, Advisor: Prof. Z. Zhang Mar. 2024 – Present

- Worked on Machine Unlearning for Large Language Models
- Worked on Kolmogorov-Arnold Networks (KANs) for reachability analysis
- Implemented post-training compression of Llama 3B using SVD-based low-rank approximation
- Developed DeepONet-based preconditioning to accelerate GMRES for 3D-IC thermal simulation

Research Assistant, UCSB, Advisor: Prof. E. Vigoda Jan. 2023 – April 2023

- Worked on designing parallel and distributed MCMC algorithms for random proper vertex colorings in general graphs

Research Assistant, University of Tehran, Advisor: Prof. A. Ghadermarzi Feb. 2022 – Aug. 2022

- Studied elliptic curves over finite fields and their applications to public-key cryptography
- Analyzed ECDH, ElGamal, digital signatures, and the elliptic curve discrete logarithm problem

MASTER'S RESEARCH PROJECT

DeepONet-Based Preconditioning for 3D-IC Thermal Simulation 2025

- Developed a hybrid multigrid preconditioner using DeepONet trunk basis functions to accelerate GMRES convergence for 3D-IC thermal simulation

GRADUATE COURSE RESEARCH PROJECTS

Machine Learning On Graphs Fall 2023

- Image reconstruction from fMRI data Using Graph Representation and Diffusion Models

Information Retrieval and Web Search Fall 2023

- Adapting a Discriminative Text Encoder for Dense Retrieval via Contrastive Span Prediction for QA Datasets

Automated Verification Spring 2023

- Modeled and verified WireGuard security properties using Tamarin Prover

Software Engineering Spring 2024

- Integrated AFL++ fuzzing into a GitHub Actions CI/CD pipeline for automated vulnerability detection

Advanced Topics in Security Fall 2023

- Graph Neural Network Approach to Classifying Vulnerabilities in Binary Files via Control Flow Graph Analysis

SELECTED PROJECTS

Deep Generative Models

- Implemented a GAN, DCGAN, and WGAN model to reproduce hand-written digits using MNIST dataset
- Implemented a convolutional autoencoder to reproduce color images using CIFAR-10 dataset

Language/Computer Vision Models

- Implemented an object detection model by developing non-max suppression on anchor boxes in YOLO model
- Implemented a sentence classification model using GloVe embedding and LSTM layers for adding emojis to sentences

Grapevine Leaves Classification

- Implemented multiple deep machine learning models for grapevine leaf classification using a transfer learning approach based on pre trained CNNs, namely MobileNetV2, ResNet50, EfficientNetB3, and InceptionNetV3
- Implemented an autoencoder to improve model accuracy and used 10 fold cross validation for evaluation

LinkedIn Implementation

- Developed an object-oriented program that had LinkedIn functionalities with graphical interface

Classroom Object Oriented Language(COOL) Compiler

- Implemented a compiler in C++ for the Classroom Object-Oriented Language(COOL), including a syntax analyzer using Flex, a parser using Bison, and a semantic analyzer using logic inference rules

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Matlab, SQL, HTML

Frameworks / SDKs: TensorFlow, PyTorch, Keras, Pandas, NumPy, SciPy, scikit-learn, MySQL, git, Docker, AWS

SELECTED COURSES

CS 292G Quantum Computing (A)

CS292F Machine Learning on Graphs (A)

CS279 Advanced Topics in Security (A)

CS 272 Software Engineering (A)

CS265 Machine Intelligence (A)

CS293S Information Retrieval and Web Search (A)

CS 293B Foundations of Blockchains and Distributed Systems (A)

MATH 260HH Special Topics in Machine Learning (A)

TEACHING EXPERIENCE

Teaching Assistant	Advanced topics in object-oriented computing	Winter 2025
Teaching Assistant	Problem Solving with Computers	Fall 2024
Teaching Assistant	Foundations of Computer Science	Spring 2024
Teaching Assistant	Foundations of Computer Science	Winter 2024
Teaching Assistant	Linear Algebra	Fall 2020
Teaching Assistant	Fundamentals of Combinatorics	Fall 2020
Teaching Assistant	Engineering Probability and Statistics	Fall 2021
Teaching Assistant	Number Theory	Fall 2021

HONORS AND AWARDS

- University of California Santa Barbara Ph.D. Fellowship and Academic Excellence Award
- NYU Tandon School of Engineering Ph.D. Fellowship
- Ranked 280 among more than 300000 participants in Iranian Nationwide Physics and Mathematics University Entrance Exam
- Ranked 112 among more than 95000 participants in Iranian Nationwide Foreign Languages University Entrance Exam
- Semi-finalist in National Olympiad of Mathematics and Informatics
- Accepted in Iranian National Organization for Development of Exceptional Talents High Schools